

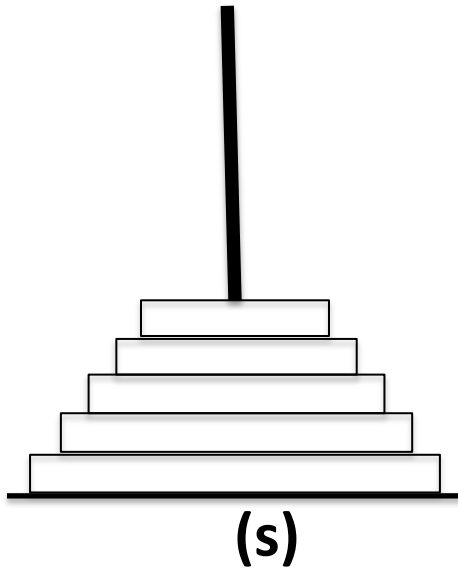
Recursive Function

(Tower of Hanoi)

What is Tower of Hanoi?

The *Towers of Hanoi* is a well-known children's game, played with three poles and a number of different-sized disks. Each disk has a hole in the center, allowing it to be stacked around any of the poles. Initially, the disks are stacked on the leftmost pole in the order of decreasing size, i.e., the larger on the bottom and the smallest on the top. The objective of the game is to transfer the disks from the leftmost pole to the rightmost pole, without ever placing a larger disk on the top of a smaller disk. Only one disk may be moved at a time and each disk must always be placed around one of the poles.

Tower of Hanoi



Task:

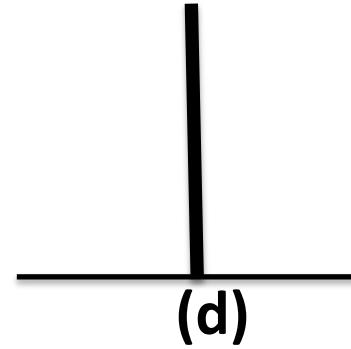
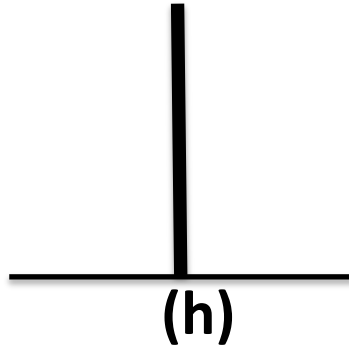
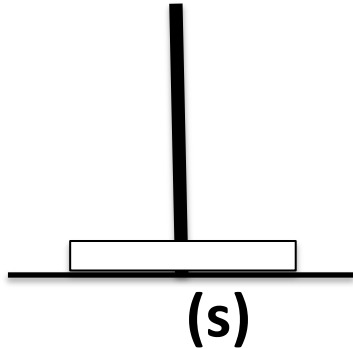
Move all disks from
pole (s) to pole (d).

Conditions:

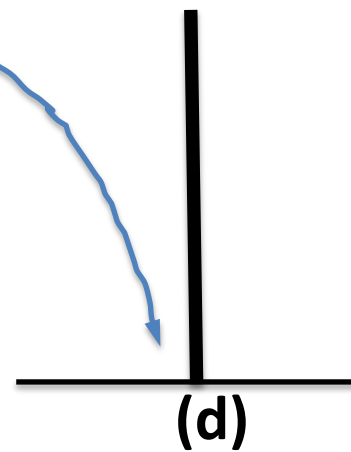
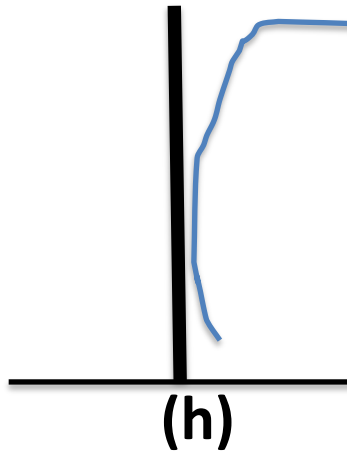
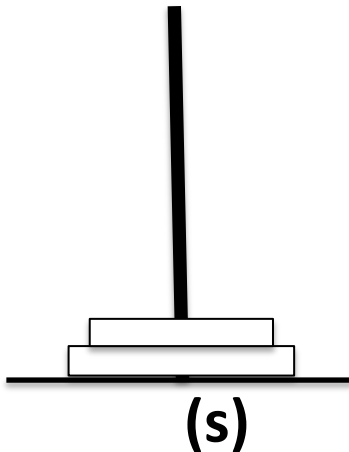
1. Only one disk can be moved at a time.
2. A larger disk cannot be placed on top of a smaller disk
3. Pole (h) can be used as a helping pole.

Tower of Hanoi -Solution

For $n = 1$

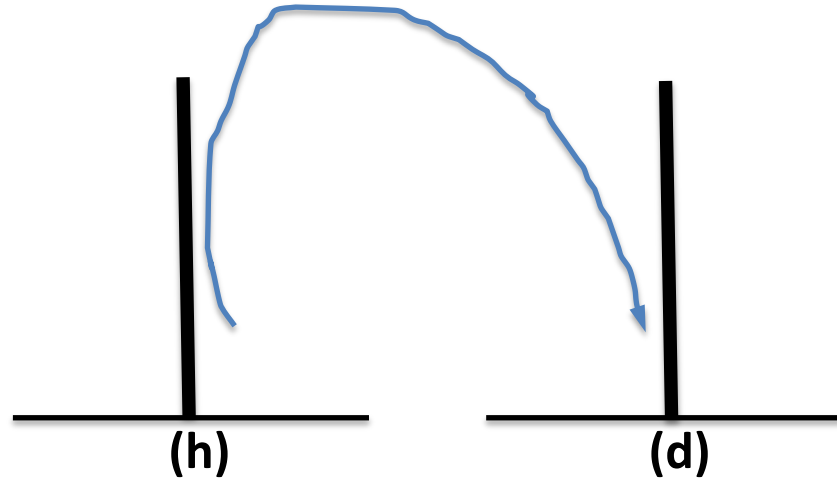
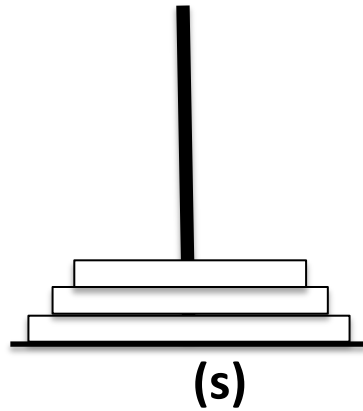


For $n = 2$

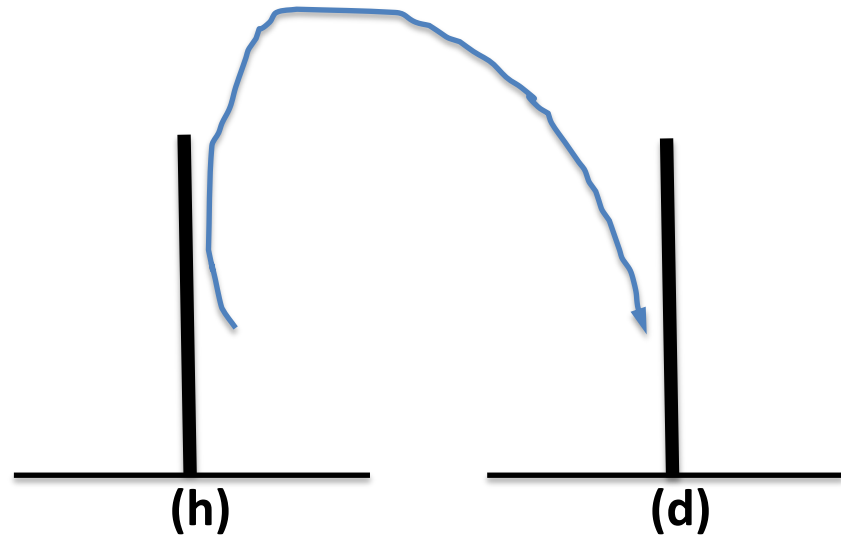
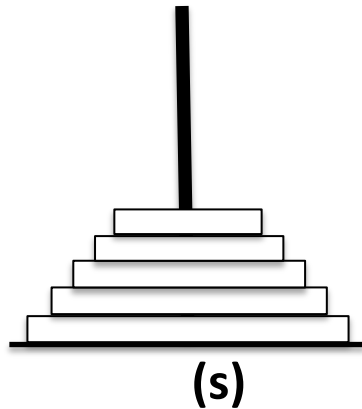


Tower of Hanoi -Solution

For $n = 3$



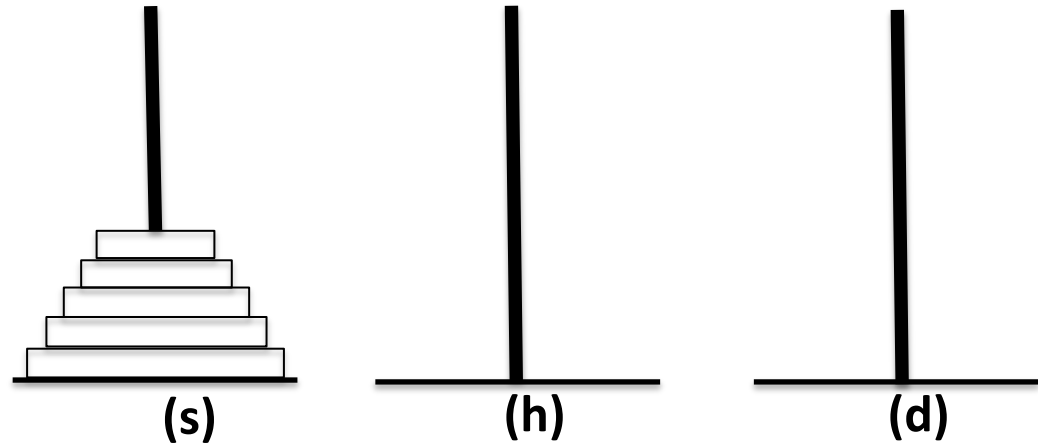
For $n = 5$



Tower of Hanoi – Program Design

Recursive Definition:

1. **Termination:** if only one disk, move from (s) to (d).
2. **Reduction:** if n ($n > 1$) disks, then follow the steps:
 - (a) move $(n-1)$ disks from (s) to (h).
 - (b) move remaining 1 disk from (s) to (d)
 - (c) move $(n-1)$ disks from (h) to pole (d)



```
void Hanoi(int n, int s, int h, int d){  
    if ( n == 1) movedisk(s, d);  
    else{  
        Hanoi(n-1, s, d, h)  
        movedisk(s, d);  
        Hanoi(n-1, h, s, d);  
    }  
}
```

Tower of Hanoi – Program

```
#include <stdio.h>
void Hanoi(int, int, int, int);
int main(){
    int n;

    printf("How many disks to move?: ");
    scanf("%d", &n);
    Hanoi(n, 1, 2, 3);
    return 0;
}

void Hanoi(int n, int s, int h, int d){
    if ( n == 1) printf("move disk from %d to %d.", s, d);
    else{
        Hanoi(n-1, s, d, h)
        printf("move disk from %d to %d.", s, d);
        Hanoi(n-1, h, s, d);
    }
}
```

Description

```
void H (n, s, h, d){
```

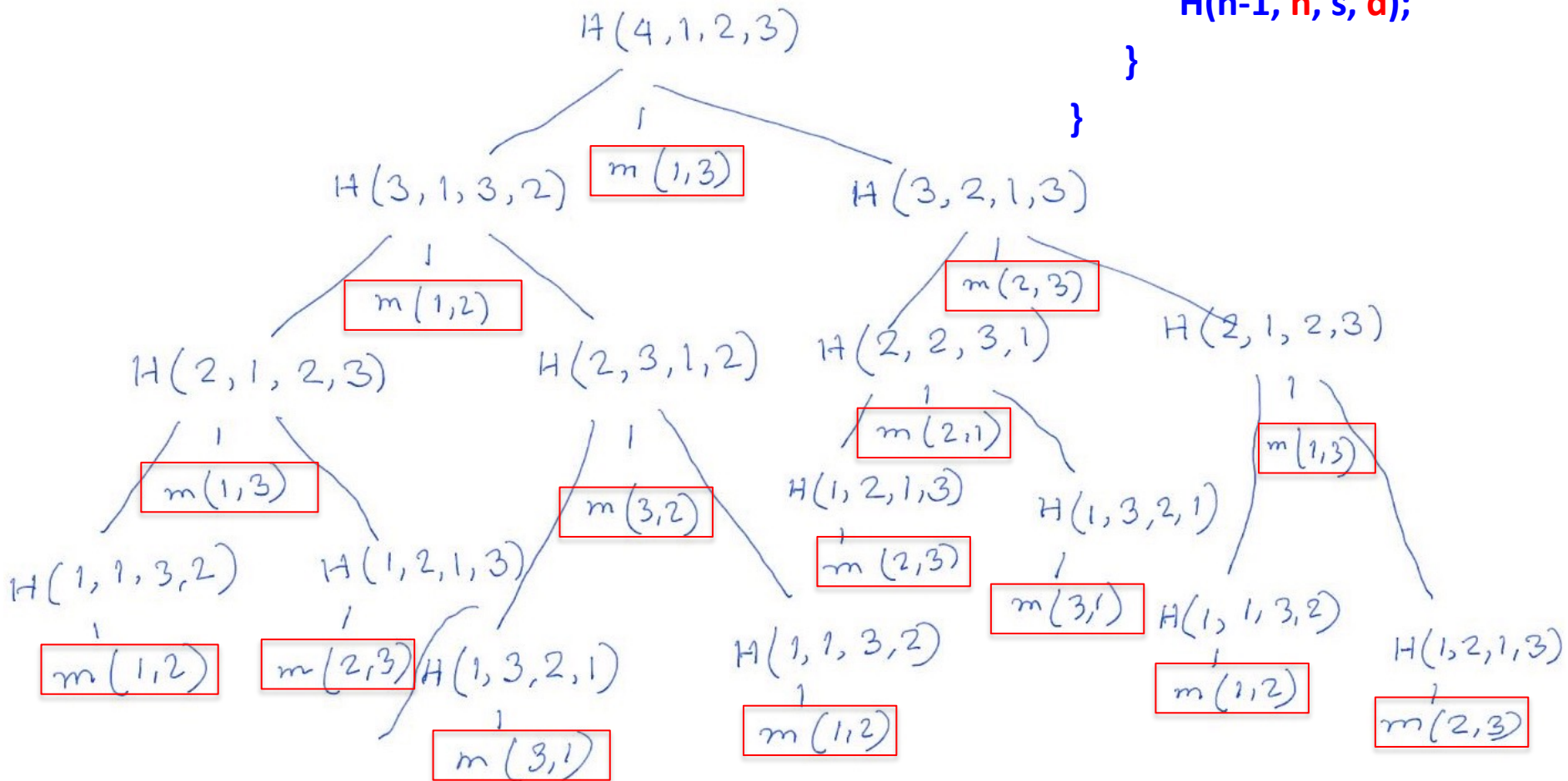
```
else{
```

 $m(s, d);$

H(n-1, h, s, d);

}

}



How many movements are required?

Number of Movements (Recursive Definition)

$$C(n) = \begin{cases} 1, & \text{if } n=1 \\ 1 + 2C(n-1), & \text{otherwise} \end{cases}$$

$$\begin{aligned} C(n) &= 1 + 2C(n-1) \\ &= 1 + 2 \{ 1 + 2C(n-2) \} \\ &= 1 + 2 + 2^2 C(n-2) \\ &= 1 + 2 + 2^2 + 2^3 C(n-3) \\ &= 1 + 2 + 2^2 + 2^3 + \dots + 2^{n-1} C(1) \\ &= 1 + 2 + 2^2 + 2^3 + \dots + 2^{n-1} \\ &= \frac{2^n - 1}{2 - 1} = 2^n - 1 \end{aligned}$$

So, no. of movements for n disks
is $= 2^n - 1$.

```
void H (n, s, h, d){  
    if ( n == 1) m(s, d);  
    else{  
        H(n-1, s, d, h)  
        m(s, d);  
        H(n-1, h, s, d);  
    }  
}
```